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A review of the application of UAV multispectral remote sensing technology in precision agriculture

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ABSTRACT

With the growing demand for precision agriculture, which requires high spatial and temporal resolution crop information, unmanned aerial vehicles (UAVs) equipped with multispectral sensors have become increasingly vital tools for agricultural management due to their real-time monitoring capabilities, flexibility, and cost-effectiveness. This paper provides a scoping review of advances in UAV-based multispectral remote sensing applications in precision agriculture, focusing on four key domains: crop growth monitoring, pest and disease identification, nutrient status assessment, and yield prediction. We offer a comprehensive analysis of the relevant literature, evaluating the principal challenges and opportunities associated with deploying this technology in the field. Our review indicates that traditional vegetation indices (e.g., NDVI, GNDVI, SAVI) have achieved mature application across diverse crops. In contrast, integrating emerging indices (e.g., TCARI, RDVI, OSAVI) with intelligent algorithms can significantly enhance monitoring accuracy and operational efficiency. In addition, we recommend that future research prioritize improvements in algorithmic precision, optimizing the data-processing workflows, and interdisciplinary collaboration to promote deeper integration of UAV-based multispectral sensing and artificial intelligence methods. Finally, we outline several practical research directions to inform and guide subsequent investigations.

1. Introduction

Precision agriculture leverages advanced technologies to optimize inputs and maximize yield while reducing environmental impacts [1]. However, traditional satellite remote sensing is limited by coarse spatial resolution [2], infrequent revisit cycles and weather-dependent [3] data gaps [4]. To overcome these drawbacks, unmanned aerial vehicles (UAVs) equipped with multispectral sensors enabling flexible, high-resolution, high-frequency field monitoring [5].

While UAV platforms share common benefits—on-demand deployment, tailored flight planning and sub-decimeter spatial detail—the choice of imaging sensor critically shapes application outcomes. RGB cameras capture fine spatial textures but lack the spectral specificity for quantitative crop-health assessment; thermal infrared imagers enable canopy-temperature and soil-moisture mapping [6]; radar penetrates clouds and vegetation for terrain and biomass analysis [7,8]; LiDAR generates high-precision 3D point clouds for structure and elevation modeling [9]; and hyperspectral systems collect hundreds of narrow bands for detailed biochemical and stress detection, albeit with heavy payloads, complex calibration and large data volumes that impede routine use [10,11]. In contrast, multispectral sensors acquire a handful of strategically selected bands (e.g., red, NIR, red-edge), striking an optimal balance: they deliver sufficient spectral relevance for robust vegetation-index computation while keeping platform weight, data throughput and processing workflows manageable [12,13].

Unmanned aerial vehicle (UAV)–based multispectral remote sensing is increasingly ubiquitous, delivering high-spatiotemporal-resolution data that reveal canopy physiology and development [14]. These data support four recurring task families in precision agriculture: (i) stress diagnostics—leveraging multitemporal reflectance and vegetation-index dynamics to estimate canopy temperature and water stress; (ii) nutrient assessment—mapping spectral-reflectance patterns to characterize crop nutrient status spatially; (iii) stand and plant phenotyping—using image segmentation and object detection to count individual plants rapidly [15]; and (iv) early-growth monitoring—integrating spectral and morphological cues from early stages to track seedling emergence [16].

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Methodologically, each task maps to specific ML/DL paradigms rather than a generic AI add-on: feature-engineered ML models (e.g., support vector machines, gradient boosting) predominate in stress and nutrient estimation, where designed inputs—vegetation indices, red-edge bands, and texture/structural descriptors—carry most of the signal.DL models learn spatial context and cross-band patterns end-to-end, with U-Net–type architectures excelling at segmentation for emergence mapping, YOLO variants at within-row detection for plant counting, and temporal networks (e.g., LSTM) at fusing multi-date imagery. Reported field results illustrate this fit—for example, YOLOv8 achieving 99.53% accuracy in tobacco plant counting and cotton seedling detection with RMSE of 2.69%—but these gains hinge on large, well-calibrated training sets that UAV multispectral platforms uniquely provide. Recent studies span rapid wheat-disease surveillance [17], broad reviews of UAV applications and plant-disease detection [18,19], row-based plant counting [20], systematic evaluation of vegetation indices for potato yield prediction [21], and assessments of practical impacts in precision agriculture [12]. Yet most reviews remain siloed by domain or method, underscoring the need for a consolidated assessment that maps task types to model choices, input features, and performance metrics in UAV-based multispectral sensing.

To address this gap, we conducted a scoping review following Arksey and O'Malley's framework [22]. We systematically searched Web of Science, Scopus, Wiley, ScienceDirect, Springer, MDPI, and PubMed using the keywords "drone," "multispectral," "remote sensing," and "agriculture" to identify relevant research articles and answer our study's research question. The main contributions of this paper are as follows:

1. A comprehensive scoping review of UAV-based multispectral remote sensing applications in precision agriculture.
2. Empirical results from relevant studies are collected and summarized to answer the research questions defined in this paper.
3. The current status of UAV multispectral remote sensing technology applications is summarized.
4. This paper presents a systematic overview of the challenges and opportunities in applying UAV multispectral remote sensing technology in agriculture.

2. UAV multispectral data acquisition and preprocessing workflow

Before feature extraction and modeling, UAV multispectral data pass through a comprehensive workflow (Fig. 1), comprising four major phases:

1. **Initial Planning and Access** *Flight Planning:* Set flight altitude, speed, image overlap (70–80% frontlap, 60–70% sidelap), and flight timing (typically within two hours of solar noon), and optimize flight routes based on wind speed, cloud cover, and other environmental conditions. *Access to Images:* Download raw images and sensor metadata (including gain, exposure time, onboard GPS/IMU logs) from the UAV after mission completion.
2. **Raw Data Preparation** *Data Export:* Export original camera files to standard image formats (e.g., TIFF) and organize filenames and timestamps. *Radiometric Calibration:* Apply dark-frame subtraction, lens-distortion correction, and empirical line calibration using in-field reflectance panels or a downwelling irradiance sensor to convert raw DN values to surface reflectance. *Multispectral Image Data:* Obtain a calibrated set of multispectral images as input for subsequent processing.
3. **Optional Fusion & Format Conversio** Spatially register and fuse multispectral images with RGB or thermal layers to enrich data dimensions; Export fused outputs or individual band stacks into desired formats (e.g., GeoTIFF, ENVI).
4. **3D Reconstruction and Analysis** *Build Dense Point Cloud:* Generate a dense point cloud from n images with 4–10 bands using feature matching and multi-view geometry. *Photo Alignment & Mesh Generation:* Perform bundle adjustment to recover camera poses; build mesh models and apply textures to reconstruct surface geometry. *Geometric Rectification:* Refine geo-referencing with ground control points (GCPs) and orthorectify using a digital surface model (DSM) to correct elevation distortions. *Image Cropping:* Crop the orthomosaic to the study area boundary, removing extraneous regions. *Information Extraction:* Derive vegetation indices, texture metrics, or 3D parameters from the orthomosaic and point cloud. *Statistical Analysis:* Conduct correlation analysis, model fitting, and accuracy assessment between extracted features and ground truth data.

Each step is precisely linked to ensure an integrated pipeline from mission planning through to final statistical outputs, and provides high-quality, reproducible inputs for feature extraction and modeling.

3. Research question

As a scoping review, this study seeks to map and synthesize the current body of literature on UAV-based multispectral remote sensing in precision agriculture. To that end, we examined the relevant research through multiple lenses and formulated the following research questions (RQs):

RQ1: Which specific agricultural challenges have been addressed using UAV-based multispectral remote sensing? This question examines how multispectral UAV platforms have been deployed to tackle key problems in crop management—such as early detection of water stress, identification of nutrient deficiencies, weed mapping, disease surveillance, and yield estimation. By synthesizing evidence across these application domains, we aim to clarify where and how UAV multispectral data have driven improvements in decision-making and to identify remaining gaps for future research.

RQ2: What methods have been employed in relevant research? This question aims to understand the current status of research methods integrated with multispectral remote sensing technology and to identify options based on scene types by comparing the scene adaptability of deep learning and machine learning.

RQ3: What features derived from UAV multispectral data have been used in agricultural research to assess crop-related characteristics? This question examines spectral indices (e.g. NDVI, GNDVI, NDRE), texture metrics (e.g. GLCM-based statistics), multi-angular reflectance measures, and other derived features from UAV multispectral imagery, focusing on their predictive performance for parameters such as yield, leaf nitrogen content, canopy water stress, and structural traits.

RQ4: What are the challenges and opportunities in utilizing UAV multispectral remote sensing in precision agriculture? This question helps to explore the current development status, limitations, and advantages of current UAV multispectral remote sensing technology, thus facilitating further exploration of relevant research avenues.

4. Method

4.1. Search strategy

To address the research questions (RQs) outlined above, we conducted a literature search using seven widely recognized academic databases: ScienceDirect, Web of Science, Scopus, SpringerLink, MDPI, PubMed, and Wiley. Two conditions were applied during the search process: (1) the keywords had to appear in the title, abstract, or keyword section of each article; (2) To characterize development trends within the field, a stepwise refinement of basic keywords relevant to the scope of this review was conducted, without imposing any time constraints. The keyword narrowing process was conducted in four distinct stages:

1. Identification phase: A literature search was conducted using the following keyword combination: ("multispectral" AND ("UAV" OR

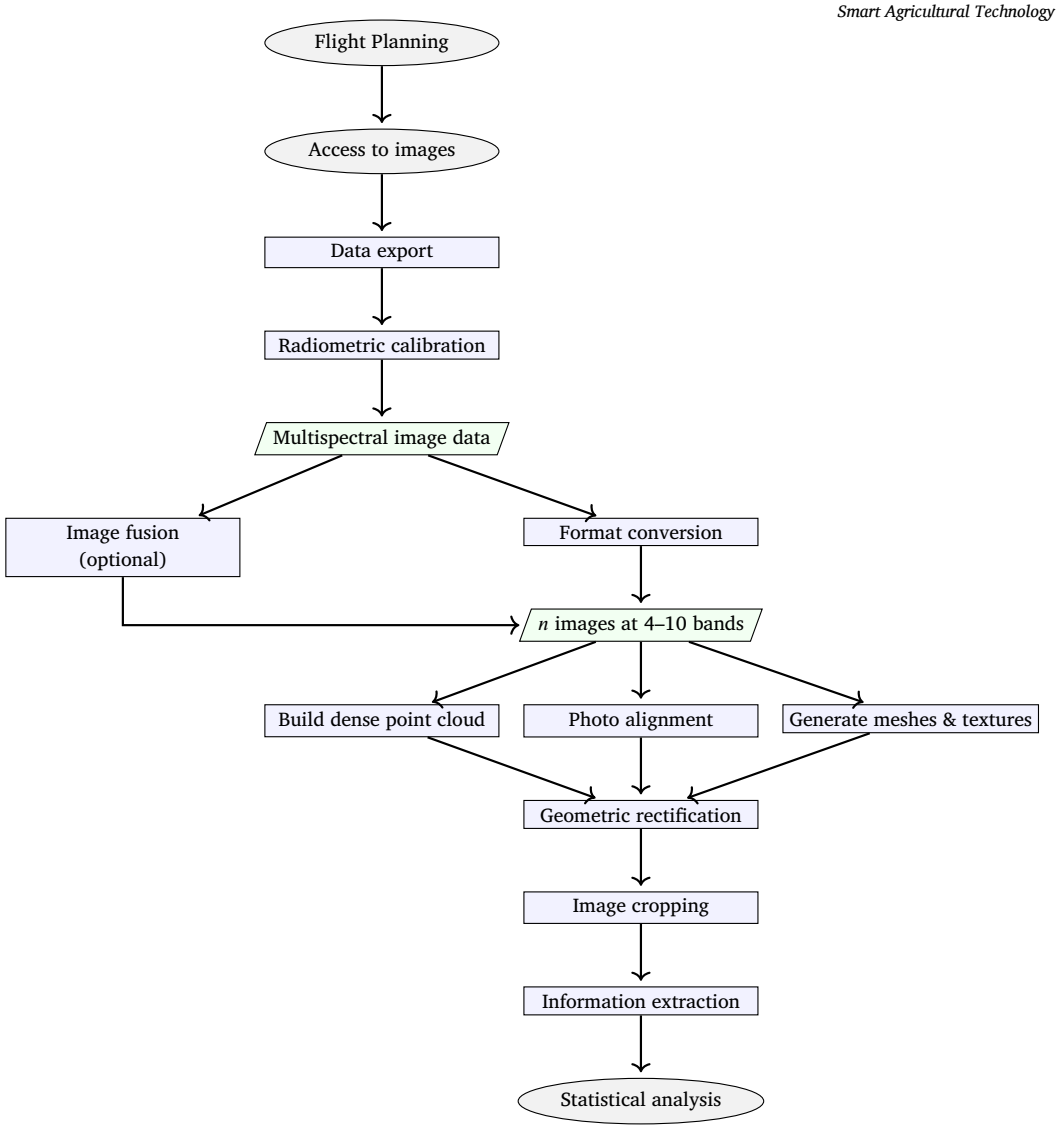

Fig. 1. UAV Multispectral Data Acquisition and Processing Workflow.

"unmanned aerial vehicle" OR "drone") AND ("remote sensing") AND ("agriculture")). The retrieved literature was then screened based on the exclusion criteria defined for each database.

2. Screening phase: The search results from the various databases were compiled, and the abstracts of the articles were reviewed to assess their relevance to the research topic. Duplicate records were identified and removed.
3. Eligibility phase: Articles closely related to the research topic were selected and retained. For example, studies using multispectral data for crop yield estimation were included, while those employing hyperspectral data or focusing on unrelated fields, such as archaeology, were excluded.
4. Data analysis phase: Articles detailing relevant methodologies or providing comprehensive descriptions of multispectral vegetation indices were selected for in-depth analysis.

To ensure that the results closely align with the paper's focus, we further refined the search string to narrow the scope. The final search string used is: (("multispectral" NOT "satellite") AND ("UAV" OR "unmanned aerial vehicle" OR "drone" OR "Unmanned Aerial Systems" OR "UAS") AND ("remote sensing") AND ("agriculture")).

After implementing the above search strategy, we found that literature published before 2015 was not included in the studies addressing RQ1-RQ3. This omission primarily results from the filtering mechanism applied during the screening stage, which indirectly excludes earlier literature. To comprehensively trace the evolution of UAV multispectral remote sensing technology in precision agriculture, this study utilizes explicitly the 'All Databases' search mode of the Web of Science database to target and supplement literature related to the development history of the technology. The relevant search results will be used to enhance the historical dimension analysis of RQ4.

4.2. Exclusion criteria

The initial search revealed numerous studies of unrelated fields (e.g., archaeology). To exclude these irrelevant studies, we conducted a thorough literature review and classified the studies according to established exclusion criteria, thereby delineating the scope of the scoping review. The exclusion criteria (EC) are as follows:

EC 1: The publication does not focus on the application of UAV multispectral remote sensing technology in agriculture.

EC 2: The publication is not a research article or conference paper.

EC 3: The publication is not in English.

EC 4: The publication is a duplicate or has been retrieved from other databases.

Table 1
Statistics on database search results.

Database	Search string	Total papers	After exclusion	After duplicate removal
ScienceDirect	Title, abstract, keywords:((“multispectral” NOT “satellite”) AND (“UAV” OR “unmanned aerial vehicle” OR “drone” OR “Unmanned Aerial Systems” OR “UAS”) AND (“remote sensing”) AND (“agriculture”))	117	76	60
Scopus	(TITLE-ABS-KEY(“remote sensing”) AND TITLE-ABS-KEY(“UAV” OR “unmanned aerial vehicle” OR “drone”OR “Unmanned Aerial Systems” OR “UAS”) AND TITLE-ABS-KEY(multispectral) AND TITLE-ABS-KEY(agriculture)) AND (LIMIT-TO(DOCTYPE, “ar”) OR “cp” OR “re”) AND LIMIT-TO(LANGUAGE, “English”)	37	10	6
Web of Science	((“multispectral” NOT “satellite”) AND (“UAV” OR “unmanned aerial vehicle” OR “drone” OR “Unmanned Aerial Systems “OR “UAS”) AND (“remote sensing”) AND (“agriculture”)) (Title) OR ((“multispectral” NOT “satellite”) AND (“UAV” OR “unmanned aerial vehicle” OR “drone” OR “Unmanned Aerial Systems” OR “UAS”) AND (“remote sensing”) AND (“agriculture”)) (Abstract) and English (Languages) and Article (Document Types)	121	72	65
Springer	((“multispectral” NOT “satellite”) AND (“UAV” OR “unmanned aerial vehicle” OR “drone” OR “Unmanned Aerial Systems” OR “UAS”) AND (“remote sensing”) AND (“agriculture”))	179	32	31
Wiley	“((“multispectral” NOT “satellite”) AND (“UAV” OR “unmanned aerial vehicle” OR “drone” OR “Unmanned Aerial Systems” OR “UAS”) AND (“remote sensing”) AND (“agriculture”))” anywhere	123	20	20
PubMed	((“multispectral” NOT “satellite”) AND (“UAV” OR “unmanned aerial vehicle” OR “drone” OR “Unmanned Aerial Systems” OR “UAS”) AND (“remote sensing”) AND (“agriculture”))	114	64	51
MDPI	((“multispectral” NOT “satellite”) AND (“UAV” OR “unmanned aerial vehicle” OR “drone” OR “Unmanned Aerial Systems” OR “UAS”) AND (“remote sensing”) AND (“agriculture”))	104	71	40
Total	-	795	345	273
Web of Science All Databases	uncrewed aerial vehicle multispectral remote sensing precision agriculture	378	-	-

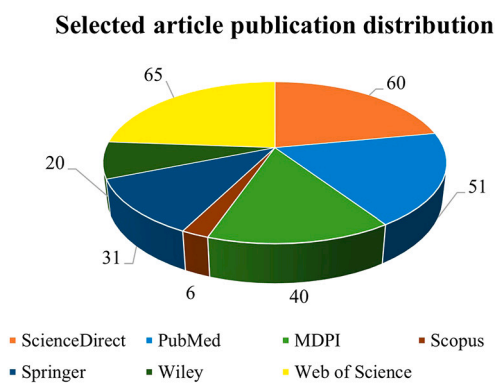

Fig. 2. Distribution of selected publications in different databases.

To accommodate differences in search options across databases, we used similar search strings as those presented in Table 1. The final number of publications selected to address RQ1-RQ3 was 273, supplemented by 378 publications used to answer RQ4, which were retrieved from the Web of Science database.

5. Results

Fig. 2 shows the distribution of the 273 records remaining after duplicate removal across seven databases. Web of Science contributed the largest share (65 articles, 23.8%), followed by ScienceDirect and PubMed (60 each, 21.9%), MDPI (40, 14.6%), SpringerLink (30, 11.0%), Wiley (22, 7.3%), and Scopus (6, 2.2%). The relatively small yield from Scopus reflects its narrower coverage of UAV multispectral remote sensing literature compared to the other platforms. All overlapping records were identified and excluded before title-and-abstract screening to avoid duplication.

To establish a clear navigational frame for the Results, we first present Fig. 3, which summarizes the Feature–Method–Task (FMT) roadmap. The figure emphasizes that what is measured from UAV multispectral data (**RQ3**) *conditions* what can be modeled (**RQ2**), which in turn *determines* where methods are most effective and deployable (**RQ1**). The arrows indicate dominant—rather than exclusive—pipelines reported in the literature and mirror the organization of the subsections that follow, including task-wise evidence and representative metrics (e.g., R^2, RMSE, OA, F1, mAP).

5.1. Which specific agricultural challenges have been addressed using UAV-based multispectral remote sensing?

Research on the application of UAV remote sensing technology in agriculture is highly diverse, but the existing literature is fragmented and lacks a systematic structure. To integrate this fragmented knowledge and address the research questions posed in this paper, we categorized the search results according to application scenarios. The classification framework can be seen in Fig. 4.

1. Growth vitality assessment
   Growth vigor includes key indicators that directly characterize the physiological status of crops, such as leaf area index (LAI), chlorophyll content, seedling emergence, failure rate, above-ground biomass, and canopy cover. These parameters provide essential data for real-time monitoring by quantifying crop health, growth dynamics, and environmental adaptation. The dynamics of crop vigor are monitored throughout the reproductive cycle, from germination to pre-harvest. Seedling emergence, a key biomass parameter, directly characterizes sowing quality and early growth potential [16]; Plant damage and survival can be quantitatively assessed using multispectral imaging from UAVs [23]; Accurate assessment of vegetation cover and plant population provides a direct basis for replanting fallen plants [24]; Collapse, as a response indicator to climatic stresses, reveals the structural integrity of crops [25].
2. Identification of Pests and Plant Diseases
   Plant pests and diseases include those caused by pathogenic microorganisms and biotic stresses induced by pests [26]. Pest and disease stresses not only lead to crop yield and quality losses but can also result in regional crop failures in severe cases [27]. Therefore, early identification and control of pests and diseases are critical for ensuring crop health and improving yield [28,29].
3. Assessment and monitoring of nutritional status
   Nutrients are indispensable components that underpin both plant growth and structural development [30]. Crop nutrient status en-

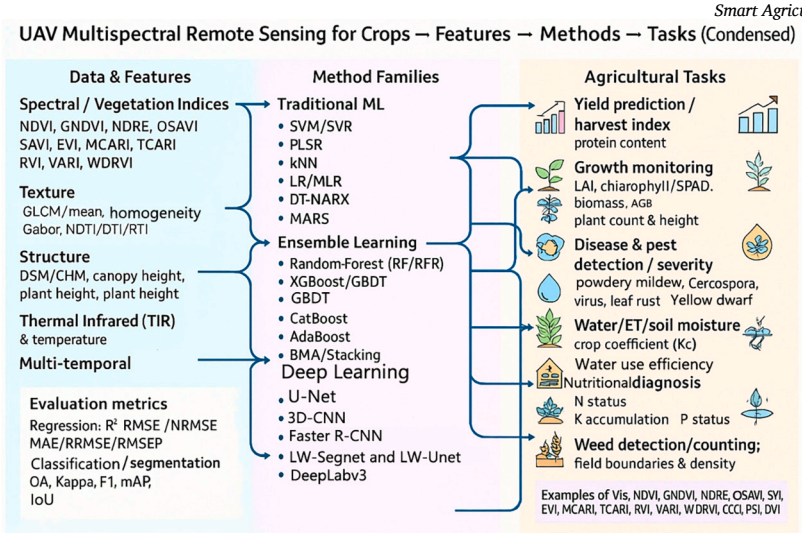

Fig. 3. Feature–Method–Task alignment for UAV multispectral remote sensing in precision agriculture.

compasses the demand, uptake, and utilization of macronutrients—particularly nitrogen, phosphorus, and potassium—as well as secondary nutrients and micronutrients by the plant throughout its growth and development [31]. Nutrient status in crops can now be assessed non-destructively using UAV-borne multispectral imagery, which links spectral responses to critical macro- and micronutrient levels. Indices such as NDVI, GNDVI and red-edge ratios correlate strongly with leaf nitrogen concentration [31,32], while blue-to-green band combinations and the Photochemical Reflectance Index (PRI) have been shown to track phosphorus deficiency by reflecting changes in energy transfer and membrane integrity [33]. Potassium status, which affects osmotic balance and enzyme activation, can be inferred through customized vegetation-stress indices and normalized difference red-edge (NDRE) metrics calibrated against Kjeldahl and flame photometry assays [34]. Calibration models—ranging from partial least squares regression to random forest—are validated via field-sampled tissue analyses, reporting typical R^2 values of 0.83 for key nutrients [35]. Integration of these multispectral approaches into decision-support systems enables optimization of fertilization regimes for N, P, K, and micronutrients, reducing overapplication, minimizing environmental impact, and promoting sustainable production [36].

4. Yield prediction
   Yield prediction is crucial in precision agriculture, as it allows farmers to accurately estimate the potential yield of different plots, enabling them to make targeted adjustments to resources such as fertilizer, irrigation, and pesticides. Additionally, protein content also influences crop yield and quality, and is therefore categorized as a yield component.
5. Monitoring of environmental resources
   Crop growth depends critically on two main environmental resources—soil and water—and UAV-borne multispectral remote sensing has been applied to both with distinct aims and methods.
   Soil Monitoring Multispectral imagery enables non-invasive mapping of soil properties and management issues. For example, normalized difference soil moisture indices (e.g., NDWI; [37]) and red-edge reflectance ratios have been used to estimate surface moisture and salinity, calibrated against in-situ probes and laboratory assays [38,39]. Weed spatial distribution, a key soil health indicator, is detected using texture-augmented vegetation indices and deep-learning classifiers, achieving classification accuracies above 85% in field trials [40–42]. Soil nutrient and texture mapping have likewise employed band-combination indices validated by soil sampling [43,44]. **Water Stress and Irrigation** UAV multispectral data support dynamic monitoring of crop water status via indices such as the photochemical reflectance index (PRI) and modified NDWI, which correlate with stomatal conductance and leaf water potential [45,46]. Evapotranspiration is modeled using artificial neural networks trained on multispectral inputs and flux-tower measurements, yielding RMSE reductions of 10%–15% compared to traditional methods ([47]). Irrigation efficiency assessments combine canopy temperature (thermal-infrared) and red-edge bands to detect over- and under-watered zones, validated against soil moisture sensors and irrigation logs [48,49].

Fig. 5 shows the distribution of the statistics of the articles selected in this paper, categorized according to the types of applications discussed in the previous section. Among these, the top three categories by number are plant pest and disease detection, growth vigor, and yield prediction, in that order. This distribution reveals that the selected articles primarily focus on areas related to plant health and productivity, reflecting the central research themes and current hot topics. Specifically, these themes emphasize how technological advancements can improve plant health and yield, addressing the needs of agricultural production and contributing to sustainable development.

5.2. What methods have been used to conduct relevant research?

UAV-based multispectral remote sensing technology in precision agriculture integrates advanced image processing techniques [50] and botanical knowledge [51] with machine learning and deep learning algorithms to construct a multilevel analysis framework. The framework uses texture indicators that can characterize the spectral and physiological properties of crops as model input features [52,53], which enriches the dimensionality of feature extraction and enables the precise monitoring and management of crop health levels and yield.

To comprehensively evaluate the efficacy of existing methods, this study classifies the approaches in the literature selected for this paper based on model complexity and feature generation methods into three categories: (1) Traditional Machine Learning, which includes non-neural network models that rely on artificial feature engineering (e.g., SVMs, decision trees, etc.); (2) Ensemble Learning, which combines base models through parallel (Bagging) or serial (Boosting) strategies to improve generalization capabilities; and (3) Deep Learning, which involves end-to-end feature learning using architectures such as Convolutional Neural Networks (CNN) and Transformers. The performance comparison of each method in crop monitoring tasks is presented in Table 2.

1. Traditional machine learning

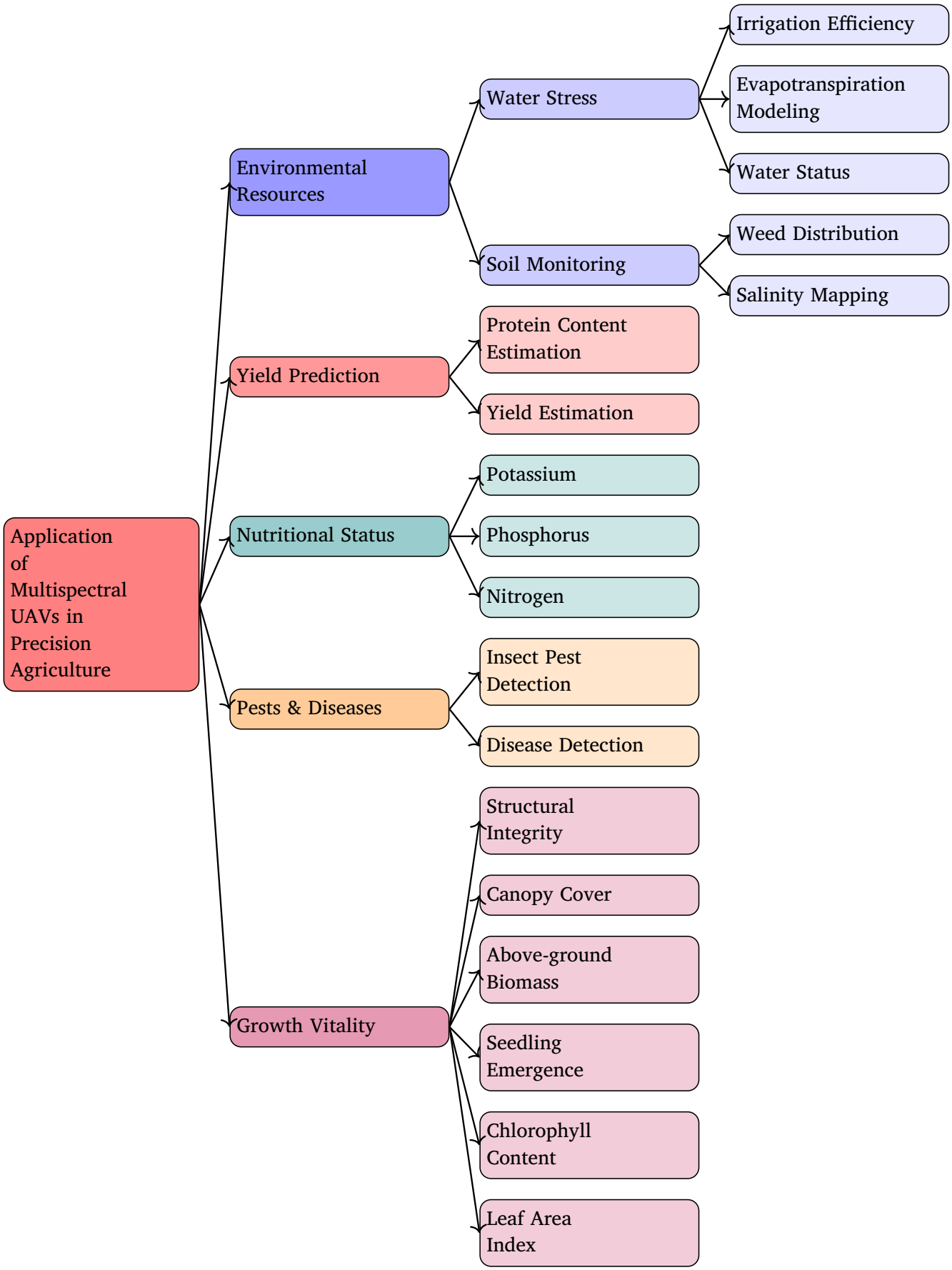

Fig. 4. Framework of UAV Multispectral Remote Sensing Applications in Precision Agriculture.

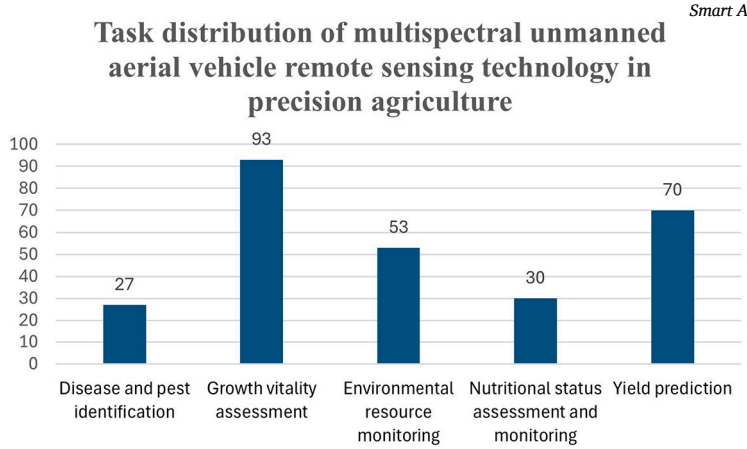

Fig. 5. Application statistics.

Table 2
Multispectral Remote Sensing Applications Analysis.

Ref.	Crop	Application	Methods[a]	Type	Metrics	Remarks
[64]	Wheat	Plant pathogens: yellow rust	DF-UNet	DL	Acc.% = 96.93	Efficient Dual Flow U-Net
[68]	Wheat	Plant pathogens: yellow rust	U-Net	DL	P = 91.30 Rec = 92.60 F = 92.00	–
[69]	Wheat	Plant pathogens: leaf rust	GA-BPNN	DL	R^2 = 92.2 RMSE = 4.88	GARI highest weight VI
[70]	Cotton	Plant pathogens: root rot	YOLOv5	DL	Acc.(%) = 70	–
[71]	Cotton	Plant pathogens: leaf blight	**MLR**/SVM/RF	ML	OA. (%) = 79%	–
[59]	Banana	Plant pathogens: Fusarium	**RF**/SVM/BPNN	EL	Acc.(%) = 97.28	Optimal 5-band RF model
[72]	Banana	Plant pathogens: Fusarium	**SVM**/RF/ANN	ML	Acc.(%) = 91.40	Spectral band inputs
[73]	Potato	Plant pathogens: Verticillium	GBM	EL	Acc.(%) = 84.00	VI + height features
[74]	Potato	Plant pathogens: late blight	**RF**/GBM/SVC/KNN	EL	Acc.(%) = 87.8	–
[75]	Wheat	Plant pathogens: scabs	**SVR**/PLSR/BPNN	ML	R^2 = 83.00	VI + TF integration
[76]	Sugarcane	Plant pathogens: white leaf	**XGB**/**RF**/DT/**KNN**	EL	Acc.(%) = 92.00	DT accuracy 91%
[77]	Citrus	Plant pathogens: HLB	**AdaBoost**/SVM /KNN/LR	EL	Acc.(%) = 99.6	Feature compression improves accuracy
[78]	Citrus	Plant pathogens: greening	SVM	ML	OA(%) = 81.75	–
[79]	Oil palm	Plant pathogens: stem rot	ANN	DL	Acc.(%) = 72.73	UAV feature extraction
[56]	Beet	Plant pathogens: Cercospora	KNN	ML	Acc.(%) = 83	–
[80]	Corn	Plant pathogens: armyworm	**RF**/MLP/NB/SVM	EL	Acc.(%) = 98.50	–
[57]	Coffee	Plant pathogens: rust	LMT	ML	P = 0.915 Rec = 0.875	–
[61]	Wheat	Plant pathogens: FHB	XGBoost	EL	Acc.(%) = 93.63 F1 = 92.93	VI + TI integration
[81]	Rubber	Plant pathogens: mildew	SVM	ML	OA(%) = 95.88 Kappa = 0.94	Spectral + texture features
[82]	Sugarcane	Plant pathogens: rust	SVM	ML	Acc.(%) = 88	–

(*continued on next page*)

Table 2 (*continued*)

Ref.	Crop	Application	Methods[a]	Type	Metrics	Remarks
[66]	Apple	Plant pathogens: blight	Mask R-CNN	DL	P = 92.8 Rec = 91.2	RVI sensitivity
[83]	Wheat	Yield prediction	RF	EL	$R^2 = 0.78$	NDVI-based model
[84]	Sugarcane	Yield prediction	NARX ANN [b]	ML	$R^2 = 0.96$	Mid-growth correlation
[67]	Rice	Yield prediction	AlexNet	DL	RMSE = 0.859 RMSPE = 14%	Multi-source inputs
[85]	Potato	Yield prediction	**SVR**/RFR	ML	$R^2 = 0.79$	GARI feature optimization
[86]	Pea	Yield prediction	**Cubist**/EN/KNN/RF	ML	$R^2 = 0.81$	Ensemble superiority
[87]	Corn	Growth: LCD/LAI/CCD	RF	EL	$R^2 = 0.91/0.97/0.97$ RMSE = 6.59/0.35/24.85	Chlorophyll metrics
[54]	Corn	Growth: chlorophyll	PLS	ML	$R^2 = 0.753$ RMSE = 2.089	10 feature selection
[88]	Coffee	Growth: maturity	RF	EL	$R^2 = 0.71$ RMSE = 11.47%	Spectral + texture
[89]	Strawberry	Growth: biomass	**ANN**/MLRRF /SVM/MARS/XGB	DL	$R^2 = 0.93$ RMSE = 7.16g	Red-edge indices
[90]	Apple	Growth: LAI	GBDT	EL	$R^2 = 0.846$ RMSE = 0.356	–
[91]	Barley	Growth: lodging	RF	EL	OA(%) = 99.7%	–
[65]	Sorghum	Growth: lodging	3D-CNN	DL	Acc.(%) = 88 P = 0.92 Rec = 0.83	–
[92]	Oilseed rape	Growth: biomass	**RF**/PLSR	EL	RMSE = 274.18 kg/ha	Texture + VI fusion
[93]	Corn	Growth: count	SVM	ML	Acc.(%) = 88.47	r = 0.98 correlation
[58]	Rice	Growth: detection	TS-DT	ML	OA(%) = 93.53 Kappa = 0.88	Texture + spectral DT
[55]	Citronella	Growth: chlorophyll	SVM	ML	$R^2 = 0.79$	DVI dominance
[35]	Wheat	Nutritional status:Nitrogen accumulation	RF	EL	$R^2 = 0.83$	Dry matter $R^2 = 0.78$
[94]	Grains	Nutritional status:Nitrogen content	R2U-Net	DL	Acc.(%) = 97.59-99.96	RGB + NIR + NDVI
[95]	Wheat	Water stress	BPNN	DL	$R^2 = 0.834$ RMSE = 0.018	Texture + VI fusion
[60]	Cotton	Water canopy	XGBoost	EL	$R^2 = 0.90$ RMSE = 0.05	VI + texture enhancement
[96]	–	Weed identification	U-Net	DL	aIoU = 0.883	Multi-spectral channels

[a] When there are multiple methods used in the same article for comparison, the best performing method is shown in bold.
[b] The NARX ANN is designed to process sequential data that contains time dependencies. Although a NARX ANN can contain multiple hidden layers, it typically differs in structure and complexity from traditional deep neural networks, and is thus classified as a conventional machine learning model.

Machine learning automatically identifies patterns from large datasets and extracts key features through a data-driven approach. Its core advantages are reflected in three key aspects: (1) interpretability (e.g., rule visualization and linear regression coefficient analysis of decision trees); (2) low computational cost (e.g., decision tree training complexity is O(n log n), making it suitable for small and medium-sized datasets); and (3) compatibility with feature engineering, as it can be integrated with artificially designed indices, such as NDVI, to enhance physical interpretability. Currently, machine learning techniques are widely used for classification, regression, and clustering tasks, covering classical algorithms such as Decision Trees (DT), Naive Bayes (NB), Support Vector Machines (SVM), and K-Nearest Neighbors (KNN). Among all the studies selected for analytical methods in this paper, machine learning was used in 35% of cases (59 out of 170) (see Table 2 for details), and these methods perform well with both structured and semi-structured data.

Traditional machine learning methods are pivotal in yield prediction, nutrient diagnostics, and disease classification due to their adaptability to small-to-medium sample sizes, interpretability, and low computational overhead. In regression tasks, Partial Least Squares Regression (PLSR) mitigates multicollinearity in hyperspectral data through latent variable decomposition, enabling stable modeling even under highly correlated bands and achieving an $R^2 = 0.753$ for maize yield prediction [54]. Support Vector Machines (SVM), combined with the Difference Vegetation Index (DVI), leverage kernel functions to capture nonlinear relationships and incorporate physiological vegetation information, yielding an

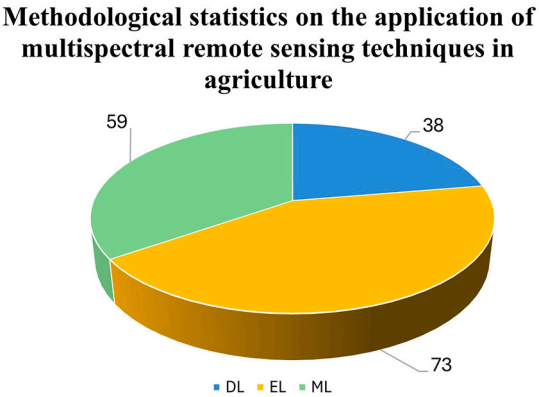

Fig. 6. Algorithmic statistics of the selected literature.

$R^2 = 0.79$ for lemongrass chlorophyll estimation [55]. The Cubist model employs a piecewise linear regression framework that balances local fitting accuracy with global model simplicity, demonstrating flexibility across growth stages and cultivars and achieving an $R^2 = 0.81$ in pea yield regression.

In classification tasks, k-Nearest Neighbors (KNN) excels with its nonparametric distance metric in scenarios with sparse samples and clear class boundaries, improving sugar beet brown spot detection accuracy to 83%[56]. Logistic Model Trees (LMT) embed spectral–textural features within a tree structure, enabling hierarchical discrimination of complex patterns and achieving an F1-score of 0.90 for coffee leaf rust detection [57], while maintaining transparent and traceable decision paths.

For object detection, the enhanced texture–spectral decision tree (TS–DT) incorporates local texture statistics at decision nodes to rapidly delineate crop contours with low computational cost, boosting plant detection accuracy to 93.5%[58]. SVM capitalizes on the sparsity of support vectors and its robust mapping in high-dimensional feature spaces to correlate maize count estimates with manual counts at $r = 0.98$, achieving precise counting without extensive annotations.

These case studies demonstrate that, under conditions of limited samples, rapid field deployment, and the necessity for model interpretability by agronomic experts, traditional machine learning—supported by concise model architectures and hand-crafted features such as vegetation indices and textural metrics—can deliver robust, high-precision, and interpretable identification and prediction in multispectral remote sensing for precision agriculture.

2. Ensemble Learning

The core objective of ensemble learning is to enhance overall predictive performance by combining multiple base models to reduce the variance (as in Bagging) or bias (as in Boosting) inherent in individual models, thereby improving accuracy, robustness, and generalization. Typical strategies include: (1) parallel ensemble methods (e.g., Random Forest), which mitigate overfitting through dual randomness in feature subsets and bootstrap sampling; and (2) sequential ensemble methods (e.g., AdaBoost), which progressively correct residuals to refine prediction accuracy. In the 42 publications analyzed in this study, ensemble learning methods appear in over 43% of cases (73 out of 170), as shown in Fig. 6. Among these, Random Forest accounts for 36% (61 out of 170) and is the most frequently employed ensemble method.

Ensemble learning (EL) methods excel in stress and nutrient diagnostics, some pest and disease monitoring, and regression-based yield or physiological indicator prediction tasks with small-to-medium sample sizes. By combining multiple weak learners in parallel or sequentially, EL approaches reduce both variance and bias simultaneously, markedly enhancing generalization and robustness. In pest and disease detection, Random Forest (RF) employs a "double-random" sampling mechanism over feature subsets and instances to smooth the spatial heterogeneity of lesions in multispectral imagery, achieving a banana fusarium wilt detection accuracy of 97.3% using five-band data [59]. In water stress and nutrient diagnosis scenarios, XGBoost leverages gradient boosting to automatically learn higher-order nonlinear interactions between multispectral vegetation indices and textural features, yielding an $R^2 = 0.90$ for cotton canopy water stress assessment [60] and a 92.93% F1-score for spectral–textural fusion detection of wheat blast [61]. Moreover, boosting algorithms (e.g., AdaBoost, Gradient Boosting) iteratively correct residuals, maintaining high-precision fitting of yield and physiological indices under limited data conditions and consistently outperforming individual models. Consequently, ensemble learning has emerged as the preferred strategy for precision agriculture remote sensing applications that demand both stability and accuracy.

3. Deep learning

Deep learning is an algorithm that learns representations of information using artificial neural networks as its architecture [62,63]. Deep learning achieves complex feature extraction and pattern recognition through multi-layer non-linear networks such as deep neural networks. The core strengths of deep learning include end-to-end learning (U-Net outputs disease masks directly from raw images [64]) and spatiotemporal modeling (Analysis of the Collapse Dynamics of Sorghum Crops Based on 3D-CNN-Fused Multiphase Imagery [65]).

Deep learning (DL) methods are particularly well suited for tasks requiring precise localization and spatiotemporal dynamic analysis, such as disease and pest lesion segmentation and crop morphological monitoring (e.g., weed detection, lodging recognition), and they have demonstrated remarkable performance in yield prediction by integrating multisource high-resolution imagery with meteorological and other temporal data. While ensemble learning (EL) has achieved excellent spectral–textural classification results in pest detection through model fusion, DL's end-to-end feature learning becomes advantageous when the objective shifts from "detecting the presence of disease" to "locating lesions and tracking their temporal evolution." For instance, U-Net's encoder automatically extracts multiscale spectral and textural features of lesions, and its decoder recovers spatial details, enabling wheat stripe rust lesion segmentation accuracy of 96.93%[64]. Mask R-CNN employs a region proposal network fused with deep features to achieve instance-level segmentation of apple fire blight with an accuracy of 91.3%–92.8%[66]. Moreover, 3D-CNNs leverage three-dimensional convolutions to jointly model spatial and temporal variations in multitemporal UAV imagery, achieving an 88% recall rate for sorghum lodging detection [65]. In multisource heterogeneous regression tasks, an AlexNet-based architecture integrating multispectral images and weather data achieved an RMSE of 0.859 for rice yield prediction, underscoring DL's strong capacity to learn and fit spatiotemporal dynamics [67]. Because deep learning can automatically capture complex interactions among spectral, textural, morphological, and temporal features within a unified framework—and perform real-time inference accelerated by GPUs—DL methods have become the technique of choice for precision agriculture remote sensing applications demanding fine-grained analysis of disease progression or crop growth dynamics.

Based on the statistics of the methods in the selected literature, as shown in Table 2, we conclude that the selection of applicable scenarios for different methods should follow three key guidelines:

1. Data Size

The deployment of deep learning models to parse high-dimensional data with end-to-end feature extraction capabilities requires large sample sizes. For example, the U-Net family of models, based on

Table 3
Hardware profile, deployment feasibility, and representative uses by learning paradigm.

Paradigm	Typical compute profile	Deployment feasibility	Representative uses
ML (e.g., SVM, DT/TS-DT)	CPU-level; 8–16 GB RAM; small models; fast training/inference	Edge devices, laptops, embedded UAV processors; real-time/near–real-time feasible	Binary/multi-class classification; small- to medium-scale regression; light-weight change detection
EL (e.g., RF, XGBoost)	Multi-core CPU; ≥32 GB RAM recommended for multi-source/texture fusion; moderate training time	Workstations for batch/offline processing; limited feasibility on low-power edge without pruning/distillation	Multi-source feature fusion; robust tabular+handcrafted-feature modeling; field mapping in post-flight
DL (e.g., U-Net, Mask R-CNN, 3D-CNN)	GPU mandatory; ≥12 GB VRAM for high-res imagery; long training; higher power draw	HPC/cloud for training; on-edge feasible only with compression/accelerators; network latency may constrain cloud inference	Pixel-level segmentation; object/lesion detection; spatio-temporal modeling and multi-modal fusion

a single high-resolution multispectral image (1336 × 2991 pixels, 5 bands), generates effective training samples through 256 × 256 pixel patching and data augmentation, achieving 96.93% accuracy in wheat stripe rust patch segmentation with an end-to-end architecture adapted for high-dimensional spectral-spatial feature resolution [64]. Similarly, Mask R-CNN combines UAV multispectral imagery (5 channels) with ground-based RGB images via transfer learning, achieving a detection accuracy of 92.8% and a segmentation IoU of 83.5% for apple fire blight detection. Its instance segmentation architecture effectively resolves the spatial distribution of infected regions in complex orchard environments [66].

For small- and medium-sized datasets, ensFemble learning or traditional machine learning methods are preferred to maximize the utilization of limited data through feature engineering. For example, SVM maintains robustness in the estimation of lemongrass chlorophyll content due to the adaptability of its kernel function to small, high-dimensional datasets ($n = 294$, $R^2 = 0.79$ [55]). XGBoost significantly improves the precision of water stress prediction in cotton canopies by integrating multispectral vegetation indices and spatio-temporal texture features, even with a limited number of samples (total sample $n = 180$, training set $n = 126$), achieving an R^2 of 0.90 [60].

2. Complexity of the Task

Pixel-level parsing tasks, such as disease segmentation and lodging detection, require the simultaneous processing of complex spatial and dynamic patterns, thus necessitating the use of deep learning models. The core challenges can be categorized as follows: (a) Fine-grained recognition under environmental disturbances—Mask R-CNN must accurately delineate fire blight patch edges within complex orchard backgrounds (branch and leaf shading, uneven illumination, and color gradients between healthy and infected areas), ultimately achieving a pixel-level segmentation accuracy of 92.8% by fusing multi-scale features through the deep receptive fields of a residual network (ResNet-101 [66]); (b) Inter-temporal 3D dynamic modeling—3D-CNN for sorghum lodging detection captures the spatio-temporal correlation between stalk bending angles (<5°) and height changes ($< 10cm$) across 11 consecutive UAV image periods (spanning 62 days), extracting spatio-temporal joint features via 3×3×3 convolutional kernels to overcome the limitations of single-phase 2D-CNN models, ultimately achieving a recall rate of 83% [65]. For regression and classification tasks of crop phenotypic indicators (e.g., yield prediction, nutritional diagnosis), ensemble learning or lightweight machine learning models are often more advantageous for handling multi-source, heterogeneous data. For instance, in fusing UAV multispectral imagery with ground sensor time-series data, Random Forest achieves an R^2 of 0.91 in maize chlorophyll density estimation through a feature subspace partitioning strategy [87]. Meanwhile, the Cubist ensemble model dynamically weights RGB and multispectral data across multiple growth stages, overcoming the saturation effect of single-sensor data, and improves cross-cultivar pea yield prediction accuracy to $R^2 = 0.81$ [86].

3. Computational and Deployment Considerations

Learning paradigm decisively shapes the hardware footprint, latency budget, and field-deployment feasibility of UAV multispectral workflows. Here we synthesize the operational profiles of traditional machine learning (ML), ensemble learning (EL), and deep learning (DL), and provide practical guidance for method selectionTable 3.

Deep learning (DL) architectures such as U-Net, Mask R-CNN, and Attention R2U-Net deliver state-of-the-art performance in pixel-level disease segmentation and biochemical estimation; however, they incur the highest computational cost. Training and low-latency inference typically require modern GPUs (e.g., ≥12 GB VRAM). In one report, Mask R-CNN training on high-resolution multispectral data used an NVIDIA Quadro P5000 (16 GB) and took ~6 h for 30 epochs [66,94]. These requirements often shift deployment to cloud/HPC backends, where network latency and connectivity can constrain in-field operations.

Ensemble learning (EL) methods such as Random Forest and XGBoost typically balance accuracy and compute. They are CPU-oriented but memory-hungry, especially when fusing multispectral, texture, and multi-source features, and thus are well suited to workstation-based batch or offline processing; pushing them to low-power edge hardware remains challenging [60].

Traditional machine learning (ML) classical models (e.g., SVM, decision trees including TS-DT) have minimal hardware overhead while maintaining competitive accuracy for many classification/regression tasks. For example, TS-DT achieved 93.53% overall accuracy in early rice detection on CPU-class hardware, indicating strong suitability for embedded processors and agricultural IoT platforms where near–real-time decisions are required [58].

Model choice should jointly consider predictive performance, computational envelope, energy and latency budgets, and integration constraints (onboard UAV vs. edge device vs. cloud). DL maximizes accuracy on large, high-dimensional datasets but at high resource and energy cost; EL offers robustness under moderate resources; ML prioritizes efficiency and is the most deployment-friendly for constrained settings.

Beyond empirical, data-driven pipelines, quantitative remote sensing for UAV multispectral sensing also includes *physical*, *semi-empirical*, and *hybrid/physics-informed learning* families. A scan of our corpus shows that strictly physical and semi-empirical approaches are markedly under-represented—only 10 of 273 papers (3.7%)—despite their strong mechanistic grounding. This imbalance helps explain why the evidence base is rich in pointwise accuracy reports but thinner on interpretability, transferability, and failure-mode diagnosis. Table 4 summarizes the alignment between tasks and model families, along with typical inputs and operational trade-offs that we elaborate below.

Physical models. Radiative transfer models (RTMs) such as *PROSAIL* retrieve canopy biophysical variables (e.g., LAI, leaf chlorophyll content C_{ab}) by simulating reflectance under controlled physiology–structure–illumination conditions, while 3D RTMs (e.g., *LESS*) explicitly

Table 4
Task–model family alignment for UAV multispectral applications (inputs and operational trade-offs).

Task / Target	Model family	Typical inputs	Strengths and limitations
LAI / Cab (biochemical traits) [97]	Physical RTM (PROSAIL); 3D RTM (LESS)	MS/HS reflectance; viewing geometry; soil/structure priors	+ Mechanistic, interpretable, scenario testing; 3D reduces anisotropy bias. – Requires calibration, prior ranges; higher setup effort; sensitive to soil/illumination if misspecified.
LAI (field-scale, time series) / Yield [98]	Crop model + DA (WOFOST + EnKF; combined methods)	UAV VI/LAI time series [99] + weather + management	+ Constrains season dynamics, links to yield; robust to gaps. – Model parameterization; DA tuning; compute moderate.
ET [100], E/T [101] partitioning, Kc, WUE [102]	Energy balance (METRIC, TSEB [103])	TIR + MS, meteorology (Rn, G, u, RH); canopy/soil temperature separation	+ Process variables for irrigation decisions; interpretable. – TIR quality, flux partition assumptions; met data dependence; careful calibration needed.
LAI / Cab (operational) [104]	Semi-empirical VI–LUT (RTM-driven)	Sensor-specific VIs (red-edge, NIR), LUTs from PROSAIL [105]	+ Lightweight, practical, mitigates VI saturation. – Dependent on LUT priors; residual structural/background effects if 3D ignored.
Biochemical traits / LAI (cross-site)	Hybrid physics-informed learning (RTM-augmented ML/DL; PITL) [106]	Simulated + observed spectra; structural cues; temporal stacks	+ Better generalization; uses priors to stabilize learning. – Pipeline complexity; needs careful synthetic-real alignment.

encode row architecture, clumping, and shadowing to reduce anisotropy and background bias in row crops. Energy-balance models (*METRIC*, *TSEB*) combine UAV thermal infrared and multispectral data with meteorology to estimate ET and partition soil–canopy fluxes, supporting water-use and drought phenotyping. Crop growth models (e.g., *WOFOST*) assimilate UAV time series (VI/LAI) via ensemble Kalman filters to constrain seasonal trajectories and improve field-scale yield estimates; empirical cues can be blended when late-season VI saturation emerges. These models provide transparent parameter pathways and scenario testing, but require careful radiometric/thermal calibration, as well as ancillary data (e.g., BRDF, meteorology), in some cases, which can involve non-trivial computational requirements.

Semi-empirical models. VI–LUT schemes bridge physics and practice by generating look-up tables from RTMs and inverting LAI/C_{ab} with sensor-tailored bandsets. Optimized VI–LUTs mitigate index saturation and reduce site-specific regressions (e.g., for wheat/cotton LAI). They are simple and moderately demanding computationally, with clearer validity domains; limitations include sensitivity to prior ranges and soil/illumination settings, and residual structural effects if 3D geometry is ignored.

Why so few in our set? Typical UAV multispectral payloads (4–8 broad bands) limit parameter identifiability in RTMs (LAI, C_{ab}, C_w, C_m), making inversions more ill-posed than with hyperspectral or multi-angle data; process-based ET retrieval needs calibrated TIR and reliable meteorology; close-range canopy geometry violates 1D assumptions and often calls for 3D RTMs or multi-angle sampling; and community toolchains incentivize fast ML/DL deployment, whereas physical/semi-empirical approaches demand stricter calibration workflows (e.g., radiometric panels/DLS, BRDF handling) and domain expertise. It is worth noting that the 10/273 share reflects our *multispectral UAV* query and does not imply limited relevance of physical/semi-empirical methods to quantitative remote sensing.

5.3. What features derived from UAV multispectral data have been used in agricultural research to assess crop-related characteristics?

5.3.1. Vegetation indices

Vegetation indices serve as the "common language" linking remote sensing technologies with crop physiological status, providing an efficient, reliable, and cost-effective foundation for crop monitoring, decision support, and resource optimization. The Vegetation Index (VI) quantifies the physiological status of vegetation, such as chlorophyll content and biomass, by combining reflectance information from specific spectral bands. The mathematical expressions of VIs commonly referenced in this paper are presented in Table 6. Most existing multispectral vegetation indices are designed based on reflectance features in the visible (VIS), near-infrared (NIR), and red-edge bands, reflecting the spectral response mechanisms of vegetation: healthy plants display low reflectance in the VIS range (400–700 nm) due to chlorophyll absorption and high reflectance in the NIR range (700–1300 nm) due to scattering by multilayered cellular structures [107]. In contrast, the red-edge band (680–780 nm) is susceptible to variations in chlorophyll concentration, effectively capturing physiological changes such as nitrogen deficiency [108] and disease infiltration [109]. Consequently, VIs incorporating both NIR and red-edge bands (e.g., NDVI, NDRE) have proven to be among the most effective tools for crop monitoring.

Based on the literature summarized in Table 5, the commonly used agricultural multispectral VIs can be classified into three categories according to their design objectives and spectral characteristics:

1. Generic vegetation indices
   NDVI, EVI, and SAVI are the most widely used general-purpose indices, primarily combining visible and near-infrared (NIR) bands for large-scale vegetation monitoring.
   NDVI is a key indicator of vegetation photosynthetic activity and canopy coverage. By quantifying the difference in reflectance between the NIR and red bands, NDVI exhibits strong correlations with the leaf area index (LAI) and biomass [110]. Its applications include yield prediction (e.g., rice, R^2 = 0.75 [111]), biomass estimation (e.g., wheat, R^2 = 0.72–0.85 [108]), and pest and disease detection (e.g., wheat blast, R^2 = 0.85 [112]) as well as fertilizer evaluation [113,114]. However, NDVI tends to saturate at high vegetation densities [115] and is significantly influenced by soil background effects, limiting its utility in monitoring chlorophyll dynamics during early to mid-growth stages. It is noteworthy that the filling stage—the most stable period in the crop growth–yield relationship—is considered the most reliable stage for crop yield monitoring [116,117].
   SAVI introduces a soil adjustment factor (L, typically ranging from 0.5 to 1.0) to mitigate background interference, thereby achieving superior results in sparse vegetation environments (e.g., peanut chlorophyll inversion, R^2 = 0.89 [118]).
   In contrast, EVI incorporates the blue band to correct for atmospheric effects, significantly improving monitoring accuracy in complex environmental conditions.
2. Red-edge sensitive vegetation indices
   Vegetation indices that utilize red-edge bands, such as OSAVI and MCARI, have demonstrated substantial advantages in nutrient diagnosis and early-stage disease detection (e.g., banana blight, overall accuracy (OA) = 91.7% [109]). OSAVI replaces the conventional NIR band with a red-edge band, achieving lower RMSE values when estimating wheat LAI under powdery mildew stress. In this condition, NDVI becomes negatively correlated with LAI [119]. MCARI optimizes the sensitivity of the chlorophyll absorption feature while suppressing the interference from non-photosynthetic materials like soil, achieving a correlation coefficient (R) of 0.919 with SPAD chlorophyll measurements [120]. Furthermore, the TCARI/OSAVI composite index significantly improves the accuracy of total nitro-

Table 5
Vegetation Indices Applications in Precision Agriculture.

Ref.	Crop	Application Scenario	VIs	Metrics	Remarks
[124]	Bean	Production forecast	NDVI, GNDVI, SAVI, DVI, EVI, TVI	$0.543 > R > 0.47$	–
[111]	Rice	Production forecast	NDVI, VARI	$R^2 = 0.75$	VIs with red-edge (720 nm) and NIR (800 nm) bands show better prediction at high canopy density
[125]	Rice	Production forecast	BWDRVI, GBNDVI et al.	$R^2 = 0.56–0.83$	Yield prediction via AGB and LAI estimation
[141]	Corn	Production forecast	MTCI, CIre, RTVI, mNDVI	$R^2 = 0.92$	All selected indices show correlations ≥0.85
[142]	Oilseed rape	Winter crop damage	OSAVI, NDVI	OA = 95%, F1 = 0.87	–
[131]	Cotton	Hail damage classification	HGVI, RVI	Kappa > 0.85	HGVI (inverse of RVI) effective for damage grading
[120]	Cotton	Leaf area index	WDRVI	R = 0.959	Strong correlation with ground-truth LAI (R = 0.951)
[143]	Sorghum	Leaf area index	NDVI	$R^2 = 0.91$	Highest correlation with LAI among tested indices
[144]	Oilseed rape	Flower count estimation	NGRDI, RGRI, MGRVI	R = 0.91	Combined with NDSI(944,758) achieves $R^2 = 0.95$
[145]	Wheat	Canopy coverage	TVI, NDRE	$R^2 = 0.96–0.97$	RedEdge-M imagery shows strong potential
[146]	Rice	Aboveground biomass	OSAVI	$R^2 = 0.63–0.78$	Best performance in pre-heading growth stages
[118]	Peanut	Chlorophyll content	NDVI, GNDVI	$R^2 = 0.86–0.89$	RVI and MCARI also show good performance
[147]	Wheat	LAI & Chlorophyll	NDVI, SAVI et al.	Acc > 87%	Investigation under different irrigation levels
[119]	Wheat	LAI estimation	GRVI, OSAVIREG	$R^2 = 0.71$	Integration of texture features improves accuracy
[108]	Wheat	Multi-parameter assessment	NDVI, REIP	$R^2 = 0.58–0.94$	NDVI best for biomass, REIP for protein content
[121]	Corn	Total nitrogen	TCARI/OSAVI	$R^2 = 0.76$	Random Forest with index ratio performs best
[148]	Rice	Potassium accumulation	RDVI	$R^2 = 0.72–0.80$	Feature fusion enhances SMLR model
[149]	Rice	Canopy nitrogen	NDRE	r = 0.80	–
[150]	Wheat	Leaf nitrogen	RDVI, SAVI	$R^2 = 0.73$	RF achieves best performance ($R^2 = 0.79$)
[126]	Cotton	Spray deposition	GNDVI et al.	–	SCCI shows strongest correlation with deposition
[151]	Banana	Planted area mapping	NDVI et al.	-	Random Forest classification method
[122]	Corn	Crop coefficient	TCARI,[a] RDVI	$R^2 = 0.68–0.80$	Red-edge based TCARI calculation
[112]	Wheat	Fusarium head blight	NDVI	$R^2 = 0.85$	Strongest correlation with disease index (DI)
[152]	Rice	Bacterial blight	NDVI	R > 0.9	NIR band normalization shows highest correlation
[109]	Banana	Fusarium wilt	CI RE	OA = 91.7%	Red-edge containing VIs outperform others
[153]	Wheat	Yellow rust	RVI, NDVI et al.	Acc = 89.3%	NIR and red bands most significant for detection

[a] The formula for calculating TCARI here is $\mathrm{TCARI} = 3\left[(\mathrm{RE}-\mathrm{R}) - 0.2(\mathrm{RE}-\mathrm{G})\frac{\mathrm{RE}}{\mathrm{R}}\right]$.

Table 6
Vegetation Indices and Formulas.

Vegetation Index	Abbreviation	Formula
Normalized Difference Vegetation Index	NDVI	$\mathrm{NDVI} = \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+\mathrm{R}}$
Green Normalized Difference Vegetation Index	GNDVI	$\mathrm{GNDVI} = \frac{\mathrm{NIR}-\mathrm{G}}{\mathrm{NIR}+\mathrm{G}}$
Soil-Adjusted Vegetation Index	SAVI	$\mathrm{SAVI} = (1+L) \times \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+\mathrm{R}+L}$
Enhanced Vegetation Index	EVI	$\mathrm{EVI} = 2.5 \times \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+C1\times\mathrm{R}+C2\times\mathrm{B}+L}$
Ratio Vegetation Index	RVI	$\mathrm{RVI} = \frac{\mathrm{NIR}}{\mathrm{R}}$
Triangular Vegetation Index	TVI	$\mathrm{TVI} = \sqrt{(\mathrm{NIR}-\mathrm{G})(\mathrm{NIR}+\mathrm{G}+k)}$
Visible Atmospherically Resistant Index	VARI	$\mathrm{VARI} = \frac{\mathrm{G}-R}{\mathrm{G}+\mathrm{R}-\mathrm{B}}$
Blue-Wide Dynamic Range Vegetation Index	BWDRVI	$\mathrm{BWDVI} = \frac{0.1NIR-\mathrm{B}}{0.1NIR+\mathrm{B}}$
Green-Blue Normalized Difference Vegetation Index	GBNDVI	$\mathrm{GBNDVI} = \frac{\mathrm{NIR}-(\mathrm{G}+\mathrm{B})}{\mathrm{NIR}+(\mathrm{G}+\mathrm{B})}$
Green-Red Normalized Difference Vegetation Index	GRNDVI	$\mathrm{GRNDVI} = \frac{\mathrm{NIR}-(\mathrm{G}+\mathrm{R})}{\mathrm{NIR}+(\mathrm{G}+\mathrm{R})}$
Pan Normalized Difference Vegetation Index	PNDVI	$\mathrm{PNDVI} = \frac{\mathrm{NIR}-(\mathrm{G}+\mathrm{R}+\mathrm{B})}{\mathrm{NIR}+(\mathrm{G}+\mathrm{R}+\mathrm{B})}$
Wide Dynamic Range Vegetation Index	WDRVI	$\mathrm{WDRVI} = \frac{a\times\mathrm{NIR}-b\times\mathrm{R}}{a\times\mathrm{NIR}+b\times\mathrm{R}}$
Meris Terrestrial Chlorophyll Index	MTCI	$\mathrm{MTCI} = \frac{\mathrm{NIR}-\mathrm{RE}}{\mathrm{RE}-\mathrm{R}}$
Red-edge Chlorophyll Index	$\mathrm{CI_{RE}}$	$\mathrm{CI_{RE}} = \frac{\mathrm{NIR}}{\mathrm{R}} - 1$
Red-edge Triangular Vegetation Index	RTVI	$\mathrm{RTVI} = 100 \times (\mathrm{NIR}-\mathrm{RE}) - 10 \times (\mathrm{NIR}-\mathrm{G})$
Modified Normalized Difference Vegetation Index	mNDVI	$\mathrm{mNDVI} = \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+\mathrm{R}-2\mathrm{B}}$
Hail Vegetation Index	HGVI	$\mathrm{HGVI} = \frac{\mathrm{R}}{\mathrm{NIR}}$
Normalized Green-Red Difference Index	NGRDI	$\mathrm{NGRDI} = \frac{\mathrm{G}-\mathrm{R}}{\mathrm{G}+\mathrm{R}}$
Red-Green Ratio Index	RGRI	$\mathrm{RGRI} = \frac{\mathrm{R}}{\mathrm{G}}$
Modified Green Red Vegetation Index	MGRVI	$\mathrm{MGRVI} = \frac{\mathrm{G}^2-\mathrm{R}^2}{\mathrm{G}^2+\mathrm{R}^2}$
Normalized Difference Red Edge Index	NDRE	$\mathrm{NDRE} = \frac{\mathrm{NIR}-\mathrm{RE}}{\mathrm{NIR}+\mathrm{RE}}$
Red Edge Inflection Point	REIP	–
Green ratio vegetation index	GRVI	$\mathrm{GRVI} = \frac{\mathrm{NIR}}{\mathrm{G}}$
Enhanced vegetation index	EVI2	$\mathrm{EVI2} = 2.5 \times \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+6\mathrm{R}-7.5\mathrm{B}+1}$
Simplified modified canopy chlorophyll content index	SCCI	$\mathrm{SCCI} = \frac{Ln\mathrm{RE}}{\mathrm{NDVI}}$
Renormalized difference vegetation index	RDVI	$\mathrm{RDVI} = \frac{(\mathrm{B}-\mathrm{G})}{(\mathrm{B}+\mathrm{G})^{0.5}}$
Optimized soil-adjusted vegetation index with red edge	OSAVIREG	$\mathrm{OSAVIREG} = 1.16 \times (\mathrm{NIR}-\mathrm{REG})(\mathrm{NIR}+\mathrm{REG}+0.16)$
Transformed chlorophyll absorption and reflection index	TCARI	$\mathrm{TCARI} = 3\left[(\mathrm{NIR}-\mathrm{R}) - 0.2(\mathrm{NIR}-\mathrm{G})\frac{\mathrm{NIR}}{\mathrm{R}}\right]$

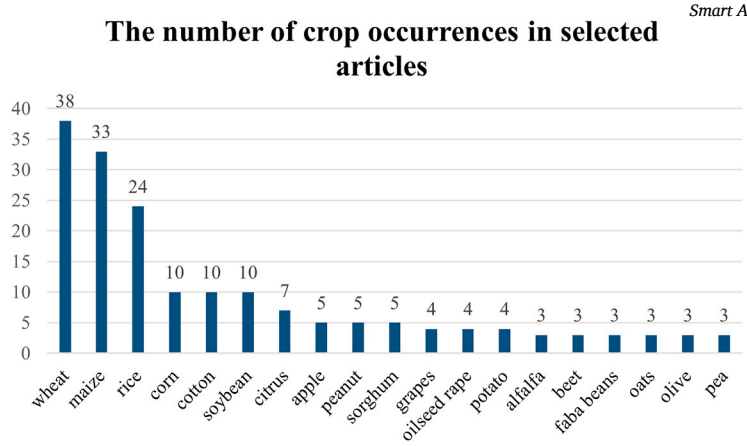

Fig. 7. Frequency of crop studies.
This crop distribution emerged organically from our systematic search (Section 4) and the application of exclusion criteria (EC 1–EC 4), rather than from an *a priori* focus on wheat.

gen content estimation in maize (R^2 = 0.76 [121]), highlighting the critical advantage of red-edge bands in diagnosing nitrogen-related anomalies. TCARI has been applied in both water stress assessments and nitrogen content estimations. However, due to the differing spectral sensitivities of water stress [122] and chlorophyll content [121], different formulas have been developed for these distinct monitoring objectives.

3. Composite functional vegetation indices
Composite functional vegetation indices are designed by coupling multispectral reflectance features with agronomic mechanisms to enhance sensitivity to crop traits under diverse conditions. Recent studies show that their design primarily reflects three directions. One direction emphasizes physiological customization, optimizing band selection to target photosynthetic or pigment-sensitive regions. For example, the green normalized difference vegetation index (GNDVI) replaces the red band with the green band to alleviate NDVI saturation, achieving high accuracy in chlorophyll estimation for peanuts (R^2 = 0.89 [118]) and supporting applications in growth monitoring [123], yield prediction [124,125], fertilizer management [126,127], and irrigation optimization [128]. This improvement leverages the principle that nutrient or water stress alters crop greenness [129], and the green band offers better discrimination than the red band [130]. Another design direction focuses on environmental interference mitigation, embedding background suppression strategies to reduce soil or moisture effects. For instance, the hail-damage green vegetation index (HGVI) computes the ratio of red to NIR reflectance to suppress soil noise, achieving a Kappa coefficient above 0.85 in cotton hailstorm assessments [131]. A further trend involves scenario-driven adaptation, integrating multi-growth-stage data or dynamic thresholds to capture phenological variability. An example is the transformed vegetation index (TVI), which has been widely applied to evaluate crop health, growth status [132], and stress levels [133,134], providing a robust foundation for precision agriculture applications. Collectively, these design tendencies highlight a shift from static formulations toward context-aware indices that integrate spectral response patterns with agronomic knowledge.
Emerging trends in multispectral inversion and vegetation index development include integrating texture features to overcome spectral mixing effects caused by canopy overlap in chlorophyll content estimation [135], and using multi-source data fusion strategies in LAI estimation [136], such as combining canopy height models with red-edge indices to reduce error [137]. The construction of emerging indices (e.g., NGRDI for nitrogen stress and disease discrimination through normalized green differences) often involves tailoring spectral bands to specific crops' photosynthetic pathways or specific monitoring objectives (e.g., disease identification or nitrogen estimation) [138–140]. Consequently, the fusion of multiple VIs is increasingly employed in practice to enhance model robustness and adaptability.

Although vegetation indices are highly effective for predicting specific agronomic parameters, they suffer from two primary limitations.

No single VI can reliably predict the same parameter across all crops. Species-specific differences in canopy structure, leaf angle distribution, and pigment composition drive these index behaviors, especially, it should be noted that water-stress monitoring rarely relies solely on multispectral VIs; in practice, thermal sensors are used to generate Water Stress Indices (WSI) and then fused with VIs (or texture features) to achieve reliable irrigation and drought assessments. From the perspective of inter-crop variability, we tallied each crop's frequency in the literature (Fig. 7) to rapidly reveal research distribution patterns and potential biases. This analysis highlights both hotspots—wheat (41 studies), maize (39) and rice (25), reflecting their global staple status and extensive existing work—and gaps, as virtually no studies address chestnut, citronella or lemon myrtle, underscoring clear opportunities for future research. Moreover, most techniques have been thoroughly validated only on these three cereals and cannot be directly transferred to crops with more complex canopy architectures or noisier backgrounds. These staples benefit from decades of satellite-based data collection, resulting in mature VI-monitoring workflows and validation protocols that facilitate straightforward adaptation to UAV platforms. Future work should deliberately target under-represented species and validate index performance under varied canopy and environmental conditions to broaden the applicability of UAV multispectral monitoring.

Moreover, nearly all reviewed studies explicitly noted that data acquisition occurred under "clear sky" and "light wind" conditions to minimize reflectance variability, indicating another practical limitation when extending models to diverse environmental or crop contexts: illumination conditions. This limitation can be mitigated by integrating a downwelling light sensor into the radiometric calibration workflow [154]. Future research should therefore expand beyond wheat, both to under-represented species and to less-ideal flight conditions, to ensure the generalizability and robustness of vegetation-index–based monitoring across the full spectrum of precision agricultural applications.

5.3.2. *Expanded feature scope*

Although vegetation indices (VIs) capture key spectral contrasts related to pigment concentration and canopy density, recent UAV multispectral research has shown that integrating orthogonal feature types—texture, spectral statistics, and multi-angular reflectance—yields sub-

stantial gains in detection accuracy, predictive power, and model robustness.

1. Texture features
   Texture descriptors quantify the spatial arrangement of pixel intensities, enabling models to detect localized anomalies—such as early pest infestations, lesion clusters, or water-stressed patches—that are often obscured by global VI averages. For example, Zheng et al. [58] extracted a suite of GLCM-based metrics (entropy, contrast, uniformity, and correlation) from red-edge and near-infrared bands and incorporated them into a Texture–Spectral Decision Tree (TS-DT). This fusion improved classification accuracy by 5.2% to 26.7%, demonstrating the effectiveness of texture–spectral integration for UAV-based crop detection.
   Similarly, Pei et al. [60] computed texture variance and second-order angular moments from multi-temporal cotton imagery and used these features to augment an XGBoost model for canopy moisture estimation; the coefficient of determination rose from $R^2 = 0.65$ to $R^2 = 0.90$. Moreover, Liu et al. [119] showed that adding texture and structural descriptors to traditional vegetation indices increased wheat leaf area index estimation R^2 by 58% under powdery mildew stress. Together, these studies highlight how texture features capture fine-scale canopy patterns—and, when fused with spectral indices, substantially enhance the accuracy and robustness of UAV multispectral analyses.
2. Multi-angular reflectance
   While traditional VIs sample reflectance at a single geometry, multi-angular approaches exploit the bidirectional reflectance distribution function (BRDF) to recover three-dimensional canopy structure—leaf angle distribution, plant height, and canopy porosity—that single-angle indices cannot capture. By acquiring imagery under varying sun–sensor geometries, one obtains angular reflectance signatures that serve as a robust "structural fingerprint." Lizarazo et al. [73] demonstrated this principle indirectly by fusing canopy height models—derived from UAV stereo imagery—with multispectral VIs in potato Verticillium detection. The combined VI + height feature set achieved 84% classification accuracy, markedly outperforming VI-only models. The height information effectively captured plant wilting and collapse, structural changes analogous to those revealed by multi-angular reflectance.
   Similarly, Vlachopoulos et al. [91] achieved an overall accuracy of 99.7% in barley lodging mapping using Random Forests on multispectral bands. Although their study did not explicitly compute angular signatures, the high accuracy reflects the model's ability to infer lodging from structural cues embedded in the data—cues that multi-angular reflectance would capture more directly by measuring reflectance variability across view angles.
   In practice, multi-angular descriptors can be extracted by computing reflectance differences or ratios between nadir and off-nadir observations for each band, or by applying principal component analysis to the angular reflectance vectors. Integrating these descriptors with standard VIs provides critical three-dimensional information—essential for tasks such as lodging assessment, disease-induced canopy collapse detection, and biomass estimation—thereby enhancing both accuracy and robustness of UAV-based crop monitoring.

By comprehensively integrating these complementary feature sets, UAV multispectral workflows can achieve significantly higher accuracy, resilience to environmental variability, and broader generalizability across diverse crops and phenological stages.

5.4. Challenges and opportunities of UAV multispectral remote sensing in precision agriculture

To investigate the development trajectory of UAV multispectral remote sensing technology, we broadened the scope of this study by conducting a secondary search within the "Web of Science All Databases" platform. This approach aimed to address research gaps identified in the initially selected literature and further explore the evolution of UAV-based multispectral applications in precision agriculture. Since its initial emergence in the academic field in 2008, UAV multispectral remote sensing has gradually gained traction, with researchers progressively expanding its applications in agricultural remote sensing.

Early research efforts primarily concentrated on three technical directions:

1. In agricultural resource management, studies established water stress models based on spectral reflectance features [155], monitored crop growth dynamics using vegetation indices such as NDVI [156,157], and explored the feasibility of estimating chlorophyll content with conventional visible light cameras [158].
2. In crop health monitoring, spectral feature-based disease identification methods were developed, including multispectral diagnosis of citrus Huanglongbing [159], olive wilt detection through a combination of hyperspectral and thermal imaging [160], and spatial mapping of weed distributions [161].
3. In land resource surveying, UAV platforms were validated for accuracy in arable land area measurements [162] and terraced landscape interpretation [163].

It is important to note that, due to the technological limitations of sensors at the time, most studies during this early stage remained at the technical validation level on small-scale field sites. A standardized monitoring protocol suitable for large-scale implementation had not yet been developed. This limitation is one reason why publications before 2015 were excluded from the selected articles analyzed in this study.

Bibliometric analysis based on the Web of Science database (using the "All Databases" strategy) reveals a continuous upward trend in both the number of publications and citation counts related to UAV multispectral remote sensing technology (as shown in Fig. 8). This indicates that research activity in this field has been increasing steadily, with growing academic interest and societal relevance. Knowledge in this domain is rapidly accumulating and diversifying, positioning UAV multispectral remote sensing as an emerging and potentially transformative direction for future agricultural research.

To reveal recent research hotspots, this study quantitatively analyzes high-frequency subject terms using CiteSpace's Burst Detection Algorithm (BDA) to identify rapidly growing research topics [164,165]. As shown in Fig. 9, the red nodes indicate keywords with significant burst intensity, and their spatio-temporal distribution reflects the dynamic evolution of research priorities. By tracing the algorithm-generated topic evolution paths, the shift of research focus can be systematically reconstructed, providing insights into the emerging frontiers of the field [166].

The results indicate that the development of UAV multispectral remote sensing technology in agriculture can be divided into four distinct phases.

(1) Phase I (2014–2016): The burst keyword was "unmanned aerial vehicle," marking the initial stage of data fusion exploration. During this period, UAV technology began to be widely applied in agricultural remote sensing scenarios, laying the groundwork for subsequent technical advances.

(2) Phase II (2015–2017): The burst keyword shifted to "data fusion." Researchers increasingly utilized parameters such as vegetation indices (e.g., NDVI), biomass, and leaf area index for detailed crop growth analysis. Efforts focused on fusing UAV and satellite data (e.g., Landsat) to enable preliminary farmland monitoring through image correction and

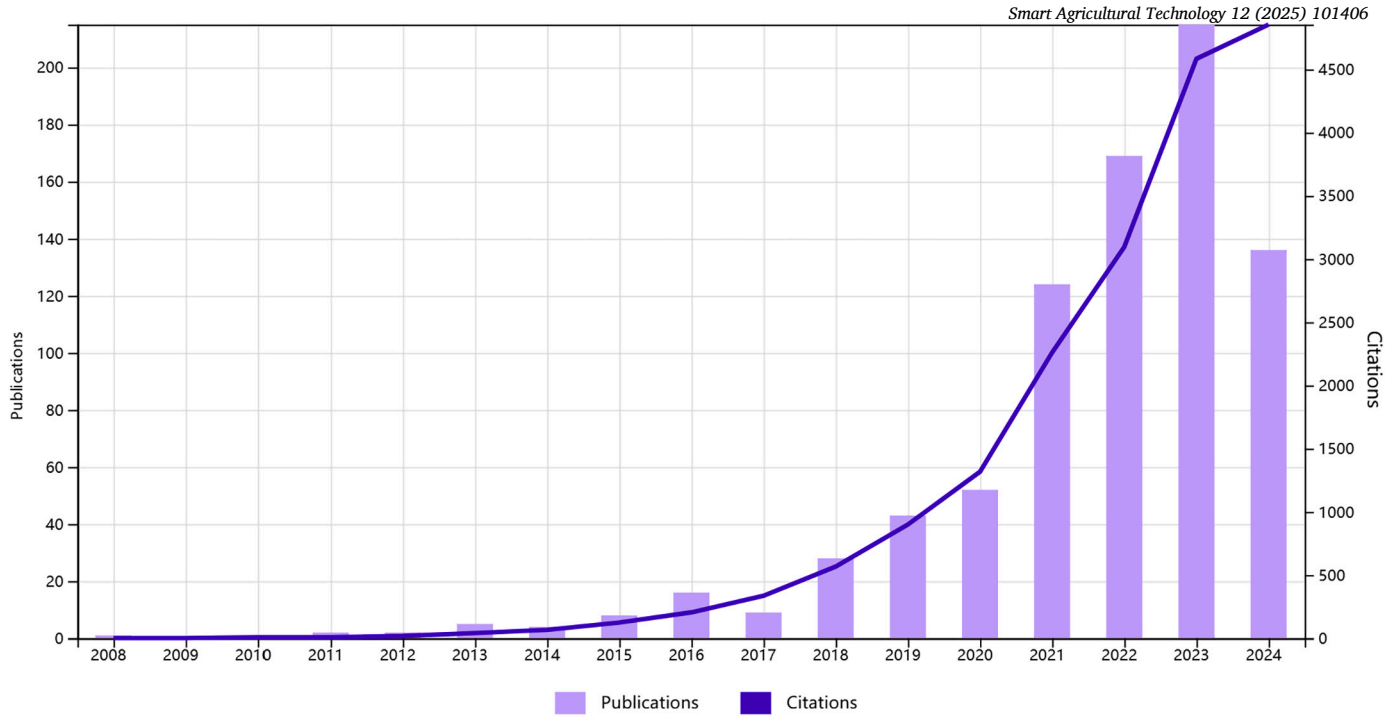

Fig. 8. Citation report for the database Web of Science.

Fig. 9. Keyword burstness results using CiteSpace.

cross-validation. Challenges related to spatial and temporal resolution discrepancies between data sources, as well as the accuracy of fusion algorithms, became the major technical bottlenecks during this phase.

(3) Phase III (2018–2019): The burst keyword was "system," indicating the deepening of data fusion techniques. At this stage, advanced UAV platforms were integrated with multispectral remote sensing to enable high-resolution crop monitoring. Through image classification and machine learning algorithms, these systems achieved precise identification of crop species (e.g., wheat, maize, cotton) and accurate estimation of key vegetation parameters such as chlorophyll content and leaf area index.

(4) Phase IV (2022–2024): With the rapid development of UAV multispectral remote sensing technology, agricultural monitoring has entered the era of "digital agriculture." This phase is characterized by integrating high spatial and temporal resolution data with intelligent analysis techniques. Digital agriculture systems leverage artificial intelligence and multispectral imagery for efficient crop monitoring and management. Models such as random forests, support vector machines, and neural networks are used to accurately monitor crop growth, while the combination of infrared reflectance and temperature data enhances disease detection capabilities. Furthermore, the application of autonomous UAVs significantly improves data acquisition efficiency, supporting critical tasks such as soil moisture detection and yield prediction within digital agriculture frameworks.

UAV multispectral remote sensing technology has significantly improved the accuracy of canopy phenotyping and stress diagnosis in precision agriculture through high-resolution dynamic monitoring and multi-band feature fusion. Nevertheless, its large-scale application continues to be constrained by several technical bottlenecks.

1. Despite the advantages of UAV technology in rapidly acquiring high-frequency, large-scale image data, the training of supervised learning models still relies heavily on high-quality and accurately labeled data samples. Currently, data annotation primarily depends on manual field sampling and interpretation, which is labor-intensive, costly, and subject to variability due to differences in expert judgment, resulting in issues of subjectivity and inconsistency. This challenge is particularly pronounced during the early stages of pest and disease outbreaks, where symptoms are often subtle and spatially heterogeneous. Under such conditions, significantly higher sampling density and broader coverage are required to obtain sufficiently diverse and representative training samples, which are essential for supporting the model's generalizability and robustness in real-world applications.
2. Multi-source remote sensing platforms—such as UAVs and sensors from different manufacturers—exhibit variations in spectral response characteristics, imaging resolution, flight altitude, and solar angle, resulting in poor comparability and reproducibility of the acquired data. This lack of standardization is particularly problematic in cross-regional and cross-crop modeling, directly limiting the transferability and generalization performance of models. In addition, traditional vegetation indices (e.g., NDVI) are prone to saturation effects in areas of dense vegetation, making them less sensitive to subtle physiological changes during high-yield periods

or the early stages of crop stress. Therefore, it is necessary to develop more universally applicable and saturation-resistant indices, along with standardized data acquisition and processing protocols, to enhance the adaptability and stability of remote sensing models.

3. Deep neural networks have demonstrated superior nonlinear modeling capabilities in tasks such as image segmentation and object recognition. However, their high-dimensional parameter space and lengthy training cycles impose substantial demands on computational resources, making them challenging to deploy directly on resource-constrained agricultural edge devices. Moreover, model tuning and training require extensive experimentation and data validation, posing a significant barrier for agricultural research teams in terms of algorithm engineering. Therefore, achieving lightweight model design and compression, such as through knowledge distillation and pruning, while maintaining model performance has become a critical research focus.
4. Integrating multimodal information—such as multispectral, thermal infrared, structured light, and environmental parameters—can enhance model diagnostic accuracy and adaptability to diverse scenarios. However, heterogeneity in temporal, spatial, and scale dimensions among different modalities can introduce noise or redundancy when combined directly. The rationality of feature selection, the complexity of fusion algorithm design, and the associated computational demands are key factors limiting the large-scale deployment of fusion models. Therefore, there is an urgent need to explore automated feature selection mechanisms and efficient end-to-end fusion strategies, such as attention mechanisms and graph neural networks.
5. Current research predominantly focuses on short-term crop performance during specific growth stages (e.g., heading, grain filling), lacking systematic monitoring across the entire crop growth cycle. This limitation hinders a comprehensive understanding of dynamic physiological changes. Additionally, significant regional variations in climate, soil, and water-nutrient conditions can lead to substantial declines in model performance when it is applied beyond the training dataset. The scarcity of cross-regional and multi-year validation data further exacerbates challenges in model generalization. Therefore, establishing standardized sample libraries and long-term observational datasets across different temporal and spatial domains is essential to support the sustainable development of this technology.
6. In practical pest and disease monitoring, different diseases often exhibit similar spectral responses or textural characteristics during early stages, leading to class confusion in mixed-disease scenarios. Moreover, existing studies often lack a systematic identification and validation of disease-specific sensitive bands or key indicators, which limits the model's ability to determine disease type and developmental stage accurately. Therefore, it is necessary to integrate plant pathology knowledge to select highly sensitive features, construct an index system for fine-grained disease classification, and incorporate expert knowledge to support model interpretability, thereby enhancing identification accuracy and reliability.

This paper argues that the interdisciplinary nature of precision agriculture requires technological innovation to rely on a multi-technology synergy framework. The existing limitations are broken through the following paths:

1. transfer learning and multimodal data synergy: extract standard features (e.g., spatial distribution patterns of spectrally sensitive bands) of UAV multispectral, radar, and thermal infrared data through transfer learning, combine them with Adversarial Domain Adaptation (ADDA) to eliminate distribution differences between sensors, and construct cross-platform feature fusion models. For example, the pre-trained convolutional layer of ImageNet is adapted to multispectral images, and the attention mechanism is used to strengthen the correlation between thermal infrared data and vegetation indices, which significantly improves the sensitivity of early stress detection.
2. Joint Optimization of Transfer Learning and Few-Shot Learning: To address the high cost of agricultural data annotation and the complexity of crop relationships, we design a Meta-Transfer Learning framework: firstly, we pre-train feature encoders on large-scale general agricultural datasets, and then we use a prototype network to achieve fast adaptation with a small number of samples in the target domain.
3. Lightweight Transfer Architecture Design: A knowledge distillation technique is adopted to transfer the multispectral response knowledge of complex models to a lightweight network. Sensitivity to key bands is retained through dynamic channel pruning, ensuring that inference speed on edge devices meets the demands of real-time monitoring.

The deep integration of agronomic knowledge and remote sensing technology is the key to optimizing the model. The risk of negative transfer in transfer learning can be mitigated by embedding plant pathology rules and constructing a multidimensional feature database that includes spectrum, texture, and environment. For example, introducing prior knowledge of diseases during the model fine-tuning stage forces the model to focus on sensitive areas through the spatial attention mechanism. Future research needs to promote data standardization and hardware adaptation further, establish cross-crop and cross-sensor pre-trained model libraries, and unify multispectral acquisition protocols (such as flight altitude and solar altitude angle) to ensure the reproducibility of transfer learning. Meanwhile, the optimization of low-altitude flight strategies and the development of cost-effective sensors are essential for promoting the widespread application of multispectral remote sensing technology in small-scale farmland and accelerating the adoption of precision agriculture technologies.

6. Discussion

6.1. Discussions based on search strategies

There may be some crossover and overlap in the categorisation of application types. For example, indicators such as crop growth status, leaf area index, and above-ground biomass also reflect, to some extent, the nutritional status of crops. Soil water use and irrigation studies can also have a direct impact on crop growth status and yield. Therefore, strictly classifying the results of these studies into a single category may lead to fragmented and incomplete information. In practice, taking these factors into account, a more integrated and cross-cutting categorization approach may be needed to better understand and assess crop growth conditions and the impact of related factors. In screening, we prioritized research articles. We discarded data articles to ensure that the selected literature provided more comprehensive background information, methodology, and analysis of results, thereby providing a deeper understanding of the application of UAV multispectral remote sensing technology in precision agriculture. However, valuable publications may have been missed during the literature search and screening process. In the literature analysis, we primarily focused on recording and analyzing UAV spectral features, which are essential for determining crop growth status, health, and environmental adaptation. Texture features can reflect the growth pattern and spatial distribution of crops, while structural features may affect photosynthetic efficiency and water utilization of crops. Including texture features, structural features, and crop variety information in the model would improve the accuracy and precision of model assessments. However, relatively few records of textural features, structural features, and related characteristics in the existing literature. Therefore, future research should consider incorporating these features into the analytical framework for a comprehensive assessment of crop growth. In the literature review, we grouped

Table 7
Comparison of UAV sensor platforms in terms of number of bands, spectral resolution, spatial resolution, revisit period, typical cost, and main applications.

Platform	Number of Bands	Spectral Resolution	Spatial Resolution	Revisit Period	Typical Cost (¥10k)	Main Applications
General Multispectral Camera	5	10–20 nm	2–5 cm	Periodic (per flight plan)	2–3	Crop health monitoring; vegetation index computation
Sentinel-2	13	10–60 nm	10–20 cm	5–10 days	1–2 (software only)	Large-scale inspection; long-term change detection
Hyperspectral Payload	270	2–5 nm	5–10 cm	On demand	20–30	Fine-scale spectral analysis; chlorophyll and nutrient estimation
Thermal-Infrared Camera	1 TIR	—	30–50 cm	On demand	5–10	Water-stress monitoring; thermal stress detection

winter wheat and other wheat varieties under the general category of 'wheat' to streamline the analysis. However, this uniform classification may obscure the differences in growth characteristics and environmental adaptability among specific wheat varieties. Therefore, future studies are encouraged to conduct more detailed classifications and analyses of individual varieties to better capture their distinct responses in the context of precision agriculture.

6.2. Discussion based on the results of this paper

There is a clear wheat-centric tendency in current research, as shown in Fig. 5, with studies related to wheat crops covering a variety of application scenarios. However, the transferability of these conclusions to other crops has not been verified, which requires a comprehensive analysis and identification of the health status of different plants at various growth stages [167]. Meanwhile, most existing experiments are limited to small-scale field observations and lack validation across different regions and multiple growing seasons, leading to a fragmented technological system.

Unmanned aerial vehicle (UAV) multispectral remote sensing holds great promise for crop monitoring, pest and disease identification, and water management in precision agriculture. Its effectiveness, however, depends on sensor characteristics and multisource data fusion. Compared to Sentinel-2–derived NDVI, UAV-NDVI is less sensitive to canopy cover variations and thus better suited for fine-scale field monitoring [168]; hyperspectral cameras offer unique advantages for leaf-level biochemical inversion and early disease detection [169]; thermal infrared sensors can generate water stress indices (WSI) but must be combined with multispectral or texture features to improve reliability. Table 7 quantitatively compares these platforms' key parameters: for small- to medium-scale monitoring, a multispectral UAV (2–3 kUSD) is preferred; for leaf-level inversion or disease alerts, a hyperspectral payload (20–30 kUSD) is required; and a thermal infrared camera (5–10 kUSD), fused with multispectral/texture features, can accurately delineate water stress. To ensure comparability between UAV and Sentinel-2 data, panel calibration and ground control point alignment should be unified; in multi-scale fusion, spatial resampling and temporal mosaicking can integrate Sentinel-2's regional trends with UAV's high-resolution details to build a hierarchical monitoring framework from individual plots to entire farm districts. Finally, sensor selection—multispectral, hyperspectral, or thermal and texture fusion—should be guided by specific application requirements.

In terms of data feature dimensions, existing studies often rely too heavily on spectral and planar texture features, while neglecting three-dimensional information related to crop morphology, spatial distribution, and stress responses. This may result in the loss of critical spatial details on crop stress.

Enhancing data processing and analysis capabilities is key to advancing the application of this technology. In plant pest and disease identification scenarios, the salient features of plant pathogenesis can generally be classified into two types: (a) In the later stages of certain plant diseases and pest infections, plants may die rapidly over a short period, significantly enhancing the spectral contrast between healthy and diseased plants, thus facilitating accurate identification. (b) In the early stages of specific pest and disease infections, the water balance and pigment structure of leaves may not yet be significantly disrupted, and anomalies in spectral reflectance are not yet prominent. In such cases, continuous tracking and dynamic analysis of spectral reflectance changes are required to capture early signals of disease development.

Regardless of the developmental stage, biotic stresses caused by diseases or pests can lead to significant yield and quality losses. Therefore, early detection and prevention of pests and diseases are far more critical than simply pursuing high detection accuracy for infected plants.

7. Conclusion

This paper provides a comprehensive analysis of 273 relevant studies retrieved from various databases. The results indicate that multispectral remote sensing technology is primarily applied to growth vitality assessment (34.1%), followed by yield prediction (25.6%), with plant disease and pest identification accounting for only 15.6%. Between 2002 and 2021, the development of multispectral technology in agriculture benefited from continuous advancements in sensors, algorithms, and numerical models. In particular, the application of UAV technology has significantly expanded the scope of multispectral remote sensing, extending from crop growth assessment to soil, moisture, and nitrogen monitoring, offering broad prospects for future development in precision agriculture.

In recent years, there has been a rapid increase in publications on UAV-based multispectral research in agriculture, especially in precision agriculture. This technology has demonstrated considerable potential to improve the efficiency and accuracy of crop management. A comprehensive analysis of the literature reveals that UAV multispectral remote sensing is widely applied in various domains, including crop growth monitoring, pest and disease identification, nutritional status assessment, and yield prediction.

When selecting methods, it is crucial to align them with the data scale and task complexity: deep learning should be prioritized for large datasets and complex pixel-level tasks, while ensemble learning or traditional machine learning is more suitable for small datasets, phenotypic prediction tasks, and scenarios constrained by limited hardware resources. Moreover, in the development and application of vegetation indices, the high sensitivity of the near-infrared and red-edge bands to crop physiological states should be fully utilized. The integration of texture features, multi-source data fusion, and customized vegetation indices will further enhance inversion accuracy and model robustness.

Future research should focus on improving algorithmic precision, optimizing data processing workflows, and exploring the synergistic use of different vegetation indices and machine learning methods to drive advancements in precision agriculture. Furthermore, interdisciplinary collaboration will play a crucial role in accelerating the application of UAV multispectral remote sensing. By integrating expertise from agricultural science, remote sensing technology, data science, and intelligent automation, UAV-based precision agriculture is expected to achieve more sustainable and efficient production models.

CRediT authorship contribution statement

Shuang Zhang: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Xiaorui Wang:** Writing – review & editing, Visualization, Software, Methodology. **Hong Lin:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis. **Yueyu Dong:** Methodology, Conceptualization. **Zhenping Qiang:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.atech.2025.101406.

Data availability

No data was used for the research described in the article.

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